Quasi-orthogonality notes

October 1, 2024

1 Preliminaries

We take everything from [1]. Here are the definitions:

1. On the sphere $S^{n-1} = \{x \in R^n : ||x|| = 1\}$, we define the **angle metric** defined as $\angle(x, y) = \arccos(x \cdot y)$. With respect to this metric, S^{n-1} has diameter π. By definition, $\angle x, -y := \angle -x, y$;
2. define the **elliptic space** P^{n-1} as the set of all unordered antipodal pairs in S^{n-1}. Here we define a similar metric, namely $\angle(x, y) = \min\{\angle(x, y), \angle(-x, y)\}$. The diameter of P^{n-1} is therefore $\pi/2$;
3. we call **spherical (elliptical) code** a subset of S^{n-1} (P^{n-1});
4. for any **finite subset** X of S^{n-1}, the **skewness of** X is the non-negative number

   $$\mu(X) := \sup\{|x \cdot y| : x \neq y \in X\}$$

   One can see that X contains antipodes if and only if $\mu(X) = 1$, and is strictly orthogonal if and only if $\mu(X) = 0$;
5. we call X **ϵ-orthogonal** if $\mu(X) < \epsilon$;
6. for any $A \in S^{n-1}$, the **ϵ-dimension** $\dim_\epsilon(A)$ of A is the larges number of elements in an ϵ-orthogonal $X \subseteq A$.

Apparently it is now easy to see the following consequences:

Lemma 1. *For $n \geq 2$, and $\epsilon \in (0, 1)$, if d is the cardinality of the largest set $Y \subset \mathbb{R}^n$ of non-zero vectors for which $y \neq u$ implies $|y \cdot u| < \epsilon||y||\,||u||$, then $d = \dim_\epsilon(S^{n-1})$.*

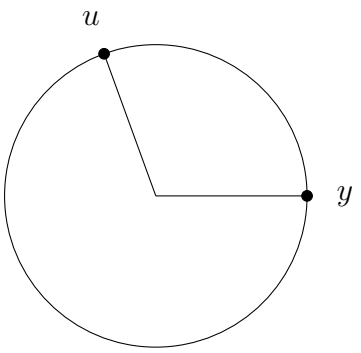
Lemma 2. *For $n \geq 2$, and $\epsilon \in (0, 1)$, if d is the cardinality of the largest set $Y \subset P^{n-1}$ of non-zero vectors for which $y \neq u$ implies $\angle(y, u) \geq \arccos(\epsilon)$, then $d = \dim_\epsilon(S^{n-1})$.*

We continue with some more definitions:

1. an **isometry** between metric spaces is a map $\omega : E \to E'$ for which holds that $h'(\omega(x), \omega(y)) = h(x, y)$, where h and h' are the distances on E and E';
2. a **segment from** x **to** y is an isometry of a real interval $[a, b]$ onto E so that $\omega(a) = x$ and $\omega(b) = y$.
3. a metric space is **excellent** if for any two points there is a segment joining them. **Assume for now that all metric spaces are excellent**.

Introducing the neighbourhoods:

1. for some $\alpha > 0$ and $s \in E$ we define the **closed α-neighbourhood** $N_\alpha(s) = \{x \in E : h(x, s) < \alpha\}$;



2. We call a neighbourhood **a cap** if we are working on a sphere (since the distance is defined as the arc distance I guess);
3. we can define the neighbourhood of a subset as the set of point neighbourhoods;

Some more general definitions:

1. for any metric space (E, h), we say that a subset $T \subseteq E$ is **s-separated** if for all $t, t' \in T$ we have that $h(t, t') > s$.
2. a family $\mathcal{F}$ of sets $F \subset E$ is a **packing of** E if for all $F \neq F'$, $F \cap F' = \varnothing$;
3. a family $\mathcal{F}$ of sets $F \subset E$ is a **covering of** E if $E = \bigcup\{F : F \in \mathcal{F}\}$;

Now a bit more specific:

1. we call T an **α-packing (covering)** if $N_\alpha(T)$ is a packing (covering). We see that T is α-separated precisely when it is an $\alpha/2$-packing.

More specifically on packings:

1. we write $\text{pack}_\alpha(E)$ for the largest number of elements in any α-packing;
2. we call an α-packing T **maximal** if $T \cup \{x\}$ is not an α-packing, if $x \notin T$;
3. we call an α-packing T **maximum** if no other packing has more elements.

Similarly for coverings:

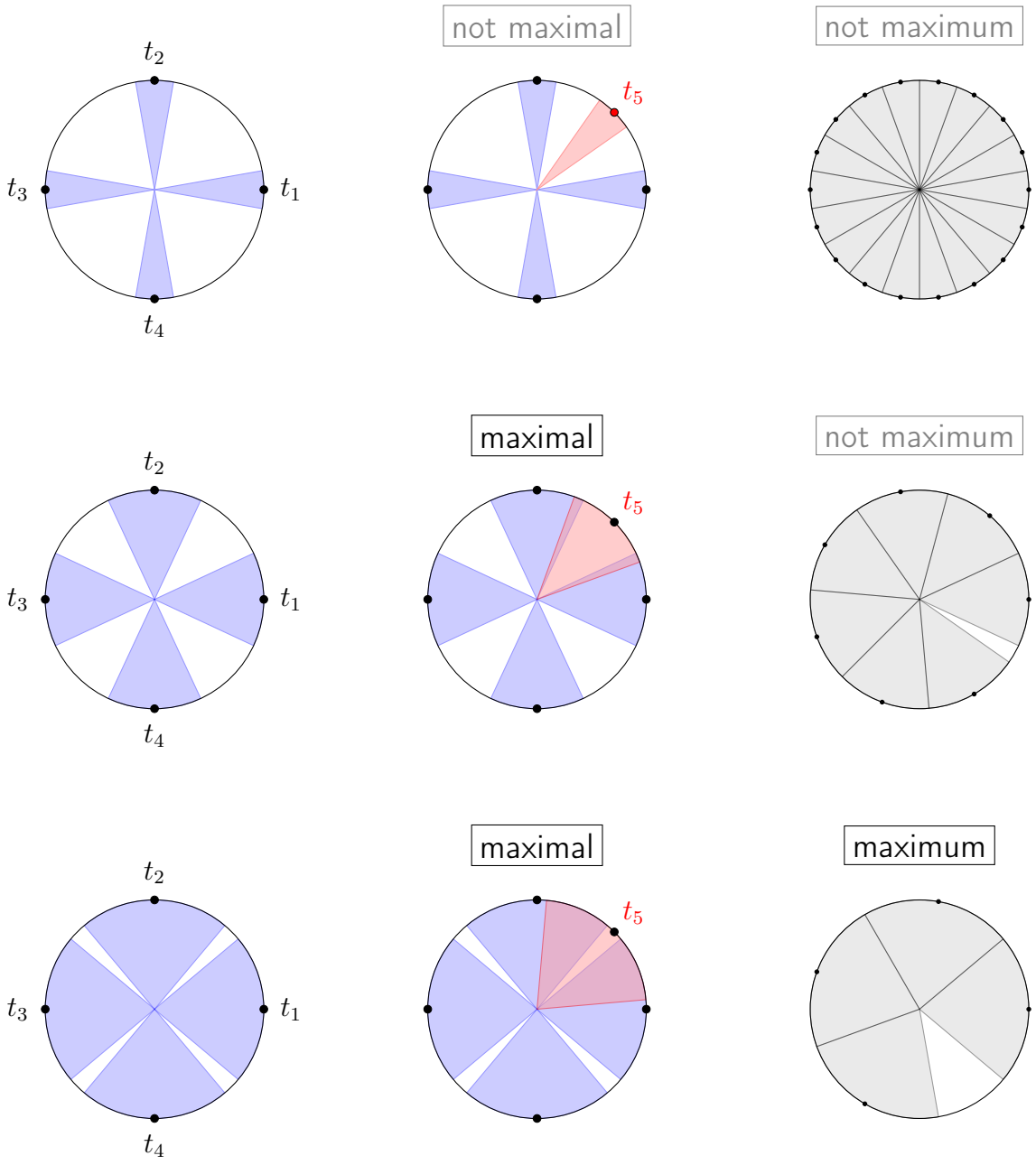
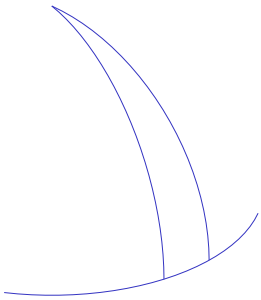
1. we write $\text{cov}_\alpha(E)$ for the smallest number of elements in any α-covering;
2. we call an α-covering T **minimal** if no subset is an $\alpha - cover$;
3. we call an α-covering T **minimum** if no other covering has fewer elements.

Now the main theorem:

Theorem 1. *For any $\epsilon \in (0, 1)$, $\dim_\epsilon(P^{n-1}) = pack_\alpha(P^{n-1})$, where $\alpha = \frac{1}{2}\arccos(\epsilon)$.*

References

1. Kainen, P. *Orthogonal dimension and tolerance* Jan. 1992.


not maximal
not maximum
maximal
not maximum
maximal
maximum
t_1
t_2
t_3
t_4
t_5